MODULE 23 · DOMAIN 3 - IMPLEMENT COMPUTER VISION SOLUTIONS

Multimodal understanding

Captions, visual question answering, alt text, objects and regions, Content Understanding fields and video segments, and standard versus pro mode.

DOMAIN 3 - Computer vision WEIGHT 10-15% MODULE 23 of 39

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A model describes what it sees. An analyzer turns what it sees into fields you can store.

Three ways to understand visual content

TOOL	BEST FOR	RETURNS
Multimodal model	Captions, visual Q&A, alt text, flexible reasoning	Text, or JSON you define per request
Content Understanding	Repeatable fields and video segments at scale	Schema-based fields, confidence, grounding
Image Analysis 4.0	Tags, objects and regions with bounding boxes	Coordinates and labels

HONEST LIMITS

Image Analysis 4.0 is deprecated and retires on 25 September 2028. Treat it as a known quantity, not a new build.

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Sending images to a model

- Send images by **URL**, as **base64**, or by **file ID**.
- **detail** is low, high or auto. **High** is needed for small text and fine detail.
- Up to **50 images** per request, up to 50 MB combined.
- Several images in one request let the model **compare** them, or caption them together.
- Structured outputs from module 12 turn descriptions into fields your code can use.

> ON THE EXAM
>
> Low detail is cheaper and faster, but a gauge reading or a serial number can become unreadable.

Concise or detailed?

CONCISE CAPTION

One sentence

- The main subject and action
- Suits thumbnails, search results, galleries
- Short enough to scan

DETAILED CAPTION

The full picture

- Objects, layout, text, relationships
- Suits catalogues and extended descriptions
- Needs high detail for small elements

ON THE EXAM

Ask for both in one call with a structured output, and let each surface use the length it needs.

Captions for several images in one call

```
content = [{"type": "input_text", "text": "Caption each product photo."}]
content += [{"type": "input_image", "image_url": url, "detail": "high"}
            for url in photo_urls]

response = client.responses.create(
    model="vision-prod",
    instructions="Describe only what is visible. Never guess brands or text "
                 "you can't read.",
    input=[{"role": "user", "content": content}],
    text={"format": {"type": "json_schema", "name": "captions",
                     "strict": True, "schema": caption_schema}})
```

ON THE EXAM

caption_schema holds a short and a long caption per image, so both lengths come from one request.

Answers grounded in visual evidence

- Instruct the model to answer **only from what's visible** in the image.
- Give it an out: **'not visible'** or **'unreadable'** when the evidence isn't there.
- Ask it to say **where** the evidence is: 'the gauge at the top left'.
- Use **high detail** for small text, dials and fine print.
- For safety-critical readings, have a person or a second check **confirm** the answer.

> THE SIGNAL
>
> A confident wrong reading usually means low detail and no permission to say 'unreadable'.

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If the pixels don't show it, the answer is 'not visible'. Grounding an image works like grounding a document.

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Alt text that follows the guidelines

- **Concise**, and focused on the image's **purpose** in its context.
- Don't start with 'image of' or 'picture of'. Screen readers already announce an image.
- **Decorative** images, like dividers, get **empty alt text**, so they're skipped.
- **Complex** images, like charts, get short alt text **plus an extended description**.
- Include essential text shown in the image, and don't guess sensitive attributes about people.

> ON THE EXAM
>
> The same photo can need different alt text on different pages. Pass the page context to the model.

Alt text or extended description?

IMAGE	ALT TEXT	EXTENDED DESCRIPTION
Product photo in a shop	'Blue waterproof walking boot, side view'	Not usually needed
Decorative divider	Empty	None
Sales chart in a report	'Bar chart: sales by region, 2026'	The values and the trend, in text
Photo of a sign with text	The sign's essential wording	Only if layout matters

ON THE EXAM

Ask the model for both fields at once, then publish the extended description only where the image needs one.

Finding where things are

- **Multimodal models** describe objects and give **approximate** positions: 'lower left'.
- **Image Analysis 4.0 Objects** returns objects with **bounding box** coordinates.
- **Dense Captions** describe up to **10 regions**, each with its bounding box.
- Captions are English only, in certain regions. The service retires on **25 September 2028**.
- Microsoft's migration guidance points to multimodal embedding models and generative models.

> ON THE EXAM
>
> Custom object detection in Image Analysis 4.0 retired in 2025. For your own object types, Microsoft points to Custom Vision.

How analyzers work

- An **analyzer** combines content extraction, AI analysis and structured output in one reusable unit.
- Works across **documents, images, audio and video**. Start from prebuilt analyzers, or build your own.
- A **field schema** names each field, its type, its description and its **method**.
- Methods: **extract** verbatim, **generate** from understanding, or **classify** into set values.
- The GA API returns **confidence scores** and **source grounding** for fields.

> ON THE EXAM
>
> Analyzers name Foundry models, not deployments. The resource maps each model to its default deployment.

A field schema for advertising video

```
"fieldSchema": {
  "description": "Extract brand presence per scene",
  "fields": {
    "brand": {
      "type": "string",
      "method": "generate",
      "description": "Brand being promoted. Include the product name."},
    "adCategory": {
      "type": "string",
      "method": "classify",
      "enum": ["Consumer goods", "Groceries", "Technology"]}}}
```

ON THE EXAM

Descriptions are instructions to the generative model. Enums keep classify results to known values.

Two stages for every video

01
Transcribe
Speech to WebVTT, with speakers.

02
Key frames
Frames that represent each shot.

03
Shots
Cut boundaries, with returnDetails.

04
Fields
A generative model fills your schema.

05
Segments
Split by your description.

06
Output
RAG-ready Markdown and JSON.

ON THE EXAM

Stage one extracts content. Stage two, fields and segments, is where a generative model spends tokens.

Whole video or custom segments?

ENABLESEGMENT: FALSE

One segment

- The whole video is analysed as one unit
- Suits compliance checks anywhere in an ad
- Suits one full-length summary

ENABLESEGMENT: TRUE

Your segments

- Describe the split in natural language
- One contentCategories object for video
- Segments from seconds to minutes long

ON THE EXAM

Segmentation uses the generative model and consumes tokens, even when no fields are defined.

What video analysis can miss

- Frames are sampled at about **one per second**, so rapid or single-frame events can be missed.
- Sampled frames are resized to **512 by 512 pixels**, so small text and distant objects can be lost.
- Only **speech** is transcribed. Music, sound effects and ambient noise are ignored.
- Set locales explicitly when a language isn't supported for multilingual transcription.
- Face identification and grouping are preview only, not in the GA API.

> THE SIGNAL
>
> 'The analyzer missed a licence plate in a fast clip' is a sampling limit, not a prompt problem.

Standard mode or pro mode?

STANDARD (SINGLE-TASK)

Extract from one file

- The default mode
- Documents, images, audio or video
- Straightforward schema extraction
- Lower cost and latency

PRO

Reason across files

- Multi-step reasoning and decisions
- Multiple input documents
- Reference data to validate against
- Documents only

HONEST LIMITS

Pro mode existed only in the 2025-05-01-preview API, which retired in July 2026. The GA API is standard.

Agentic mode (preview)

- Set **config.workflow** to **agentic** on a document analyzer (API 2026-06-01-preview).
- Builds answers from evidence: **multistep reasoning**, **calculations**, **validation**.
- Interprets complex **tables and figures** visually.
- Initial preview: **one input file** per request, which can hold related documents.
- Use it when standard extraction can't produce the answer, because it needs reasoning.

> ON THE EXAM
>
> Microsoft doesn't call agentic mode pro mode's replacement, but it's the current way to get pro-style reasoning.

Requirement to tool

REQUIREMENT	USE
Captions for a photo gallery	Multimodal model, concise caption
Answer questions about a site photo	Multimodal model, high detail, grounded
Alt text across a website	Model with page context; empty alt for decorative
Same fields from thousands of videos	Content Understanding analyzer
Split lectures into topic chapters	Custom segmentation, enableSegment true
Bounding boxes for common objects	Image Analysis 4.0 Objects (retiring 2028)
Validate totals against conditions	Agentic mode (preview)

ON THE EXAM

Flexible, one-off understanding: a model. Repeatable fields at scale: an analyzer.

> A maintenance app sends photos of pressure gauges to a multimodal model with detail set to low, asking whether each reading is above the safe limit.
>
> Answers are often confident and wrong, including for photos where the dial is blurred.

What should the team change?

A Raise the temperature so the model considers other possible readings.

B Use Image Analysis 4.0 captions to describe each gauge photo instead.

C Use high detail, and let the model say a dial is unreadable.

D Resize every photo smaller so that the dial fills more of the image.

◆ PAUSE AND DECIDE

What should the team change?

A Higher temperature.

More variety in answers, not more accuracy in reading the dial.

B Image Analysis captions.

A one-sentence caption won't report a precise needle position.

C High detail, and an 'unreadable' out.

Fine detail becomes visible, and blurred dials stop producing guesses.

D Smaller photos.

Shrinking the image loses exactly the detail the reading needs.

THE SIGNAL

Low detail plus no permission to say 'unreadable' produces confident wrong readings.

> A retailer auto-generates alt text for its website. Every image, including decorative dividers, gets a paragraph beginning 'An image of'.
>
> Screen reader users complain the pages are slow and tiring to navigate, and charts still aren't explained.

What should the team change?

A. Use Dense Captions, describing up to ten regions of every image.

B. Short alt text by purpose, empty for decorative, more for charts.

C. Set detail to high so that every paragraph becomes more accurate.

D. Use one fixed caption, 'product image', for every image on the site.

◆ PAUSE AND DECIDE

What should the team change?

A. Dense Captions everywhere.
Even more text to hear, on every image, including decorative ones.

B. Guideline-based alt text.
Short and purposeful, skips decoration, and explains charts properly.

C. High detail.
More accurate paragraphs are still paragraphs, on dividers too.

D. One fixed caption.
Tells screen reader users nothing about any image.

THE SIGNAL

Every complaint maps to a guideline: too long, 'image of', decorative noise, unexplained charts.

> A broadcaster wants its news archive searchable by story. Each hour-long bulletin should become separate chapters, one per story, each with a summary.
>
> Adverts between stories should be ignored.

What should the team configure?

A An analyzer with enableSegment true, describing one segment per story.

B An analyzer with enableSegment false, summarising each whole bulletin.

C Image Analysis 4.0 Dense Captions run on every frame of each bulletin.

D Speech transcription only, searched for the words 'next story'.

◆ PAUSE AND DECIDE

What should the team configure?

A Custom segmentation by story.

Natural-language segmentation creates story chapters and can skip adverts.

B Whole-video summary.

One summary per bulletin, with no chapters at all.

C Dense Captions per frame.

Describes still images, with no idea where stories begin or end.

D Search the transcript.

Presenters don't reliably say 'next story', and ads aren't skipped.

THE SIGNAL

'Split by meaning, in your own words' is custom segmentation with a contentCategories description.

> An insurer processes claim files that hold a claim form, an itemised invoice and the policy conditions. Standard extraction returns every field correctly.
>
> But the team also needs to know whether the invoice total matches the claim, and whether each item is covered by the conditions.

What should the team use?

A Add more fields to the standard analyzer for every invoice line item.

B Image Analysis 4.0 Objects to locate each total on the scanned pages.

C A video analyzer with segmentation to split the file into documents.

D An analyzer in agentic mode, to reason, calculate and validate.

◆ PAUSE AND DECIDE

What should the team use?

A More standard fields.

Extracts more values, but still doesn't compare or validate them.

B Image Analysis Objects.

Finds where things are, not whether numbers agree.

C Video segmentation.

For video, not document files.

D Agentic mode.

Built for calculations and validation against conditions, the pro-mode kind of problem.

THE SIGNAL

Fields already extract correctly, but the answer needs reasoning across them: pro-mode territory, now agentic mode.

Multimodal understanding in five lines

- **Models describe; analyzers produce fields.** One-off flexibility versus repeatable scale.
- **Visual grounding:** high detail, answer from the pixels, say 'unreadable' when needed.
- **Alt text:** concise and purposeful, empty for decorative, extended for complex.
- **Video:** two stages, custom segments by description, about 1 FPS at 512 pixels.
- **Pro mode** was a retired preview; **agentic mode** now reasons over documents.

ON THE EXAM

Next, module 24: responsible AI for visual content, from unsafe images to prompt injection in pictures.